

EFC Euclid DR1 Pre-Registered Benchmark Prediction

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Abstract. We register a set of frozen, falsifiable predictions for Energy-Flow Cosmology (EFC) against the forthcoming Euclid Data Release 1 (DR1, scheduled 21 October 2026). The predictions are computed through full Boltzmann integration using `hi_class` with a custom `efc_logistic` gravity model implemented in the CLASS C-code (81 lines). The benchmark parameters $B_0 = 0.02$, $M_0 = 0.06$ lie within the Horndeski stability region ($c_s^2 > 0$, $Q_S > 0$). Five observables are frozen with a SHA-256 hash, forming a coherent multi-observable signature that decouples growth and lensing in a pattern not reproduced by massive neutrinos or known systematic errors. This note serves as a publicly timestamped, immutable record of the prediction prior to data release. The prediction is fully specified prior to data release and does not rely on post-hoc parameter fitting.

1 Model summary

Energy-Flow Cosmology modifies the perturbation sector of Λ CDM through a scalar entropy field $S(a)$ that activates at late times. The background expansion is identical to Λ CDM (the “ α -gate” is closed). All EFC physics enters through three functions:

$$\mu(k, z) < 1 \quad (\text{suppressed growth}), \quad (1)$$

$$\eta(k, z) = \Phi/\Psi > 1 \quad (\text{gravitational slip}), \quad (2)$$

$$\Sigma(k, z) > 1 \quad (\text{enhanced lensing}). \quad (3)$$

The Horndeski α -functions are driven by the entropy sigmoid:

$$S(a) = [1 + \exp(-(\ln a - \ln a_t)/\Delta)]^{-1}, \quad (4)$$

$$\alpha_B(a) = B_0 \cdot \frac{dS}{d \ln a}, \quad \alpha_M(a) = M_0 \cdot S(a), \quad \alpha_T = 0. \quad (5)$$

The tensor speed constraint $\alpha_T = 0$ follows from GW170817. GR is recovered automatically at high redshift ($S \rightarrow 0$ as $a \rightarrow 0$).

2 Pipeline

The predictions are computed through the following chain:

1. EFC entropy field $S(a)$ defines $\alpha_B(a)$, $\alpha_M(a)$ (Eq. 5).
2. `hi_class` (modified CLASS with Horndeski support) solves the full Boltzmann hierarchy including perturbation-sector modifications.
3. The custom `efc_logistic` gravity model is implemented directly in the CLASS C-code (81 lines in `gravity_smg.c`), with analytical δM_*^2 computation to avoid ODE stiffness.

4. A 36-configuration parameter scan identifies the stability region ($c_s^2 > 0$, $Q_S > 0$).
5. Output: C_ℓ^{TT} , $C_\ell^{\phi\phi}$, $P(k, z)$, σ_8 , $f\sigma_8(z)$ at 8 redshifts.

The pipeline is fully reproducible from the git repository (branch: `claude/euclid-pipeline-setup-IE3e9`, 5 commits).

3 Benchmark parameters

Parameter	Value	Description
B_0	0.02	Braiding amplitude
M_0	0.06	Planck-mass running amplitude
a_t	0.30	Transition scale factor ($z_t = 2.3$)
Δ	0.30	Transition sharpness
α_K	1.0	Kineticity (stability)
α_T	0	Tensor speed (GW170817)
h	0.6736	Planck 2018 best-fit background
ω_b	0.02237	
ω_c	0.1200	
n_s	0.9649	

Table 1: Frozen benchmark parameters. EFC parameters are fixed; background follows Planck 2018.

4 Frozen predictions

Observable	Location	Shift vs Λ CDM	Direction
σ_8	global	+1.21%	↑
$P(k=0.05, z=0.5)$	k_c	+2.09%	↑
$C_\ell^{\phi\phi}(\ell=66)$	lensing	−6.01%	↓
$E_G(\ell=66)$	slip test	−3.98%	↓
$C_\ell^{TT}(\ell=30)$ ISW	temperature	+1.88%	↑

Table 2: Pre-registered predictions at benchmark $(B_0, M_0) = (0.02, 0.06)$.

4.1 Observable definition

The E_G statistic is computed from angular power spectra as

$$E_G(\ell) = \frac{C_\ell^{g\kappa}}{\beta C_\ell^{gg}}, \quad \beta = \frac{f}{b}, \quad (6)$$

using `hi_class` outputs with linear bias $b(z)$ and growth rate $f(z)$. In practice, the bias dependence cancels to leading order through the ratio $C_\ell^{g\kappa}/C_\ell^{gg}$, with residual dependence entering only via $\beta = f/b$, which is independently constrained from RSD measurements. This makes E_G the most robust test of gravitational slip.

4.2 Signal amplitude and binning

The quoted amplitudes in Table 2 correspond to intrinsic (unbinned) ℓ -space values. Euclid tomographic binning and window-function convolution are expected to reduce the signal amplitude by ~ 30 – 40% , which is accounted for in the interpretation threshold below.

4.3 Coherence property

All five observables shift in a pattern that is qualitatively distinct from known systematics:

- σ_8 and $P(k)$ increase (enhanced effective coupling at late times).
- $C_\ell^{\phi\phi}$ decreases (modified lensing kernel).
- ISW increases (time-varying potentials from $\alpha_M \neq 0$).
- E_G decreases (modified $\Sigma/(f\mu)$ ratio).

Importantly, the growth and lensing sectors respond with opposite sign—a feature not reproduced by standard extensions of Λ CDM. The signal is localized around $k_c = 0.05 h/\text{Mpc}$, providing an additional discriminator against scale-independent systematics. Massive neutrinos suppress both σ_8 and $P(k)$ simultaneously while leaving $\eta = 1$ everywhere. Galaxy bias shifts amplitudes but does not produce scale-localized features at k_c . Intrinsic alignment affects lensing but not temperature or growth. No known single systematic produces this five-observable pattern.

4.4 Kill criterion

If Euclid DR1 measures $|E_G(\ell \approx 66) - E_G^{\Lambda\text{CDM}}| < 1\%$ at 2σ precision, consistent with zero within systematic uncertainties, the EFC perturbation-sector coupling at this benchmark is falsified.

5 Integrity record

Item	Value
SHA-256 (prediction block)	dbc73797...db6e68135d2
Git branch	claude/euclid-pipeline-setup-IE3e9
Final commit	797b904
Date sealed	12 April 2026
Euclid DR1 release	21 October 2026

Table 3: Provenance record.

Full SHA-256 hash of the frozen prediction block:

dbc737979d8a773f56c620759ddded38139157fd9a2dbb321a8ffdb6e68135d2

This hash covers the benchmark parameters (Table 1), predictions (Table 2), and the `hi_class` output files. Any modification to the prediction block changes the hash.

6 Limitations

- The benchmark uses a single point in (B_0, M_0) space. A full posterior via PolyChord with Planck priors is planned but not yet computed.
- The Limber approximation introduces $\sim 5\text{--}10\%$ amplitude uncertainty at $\ell \approx 66$. Non-Limber validation is planned.
- Braiding ($B_0 > 0$) is constrained by gradient stability ($c_s^2 > 0$). The benchmark value $B_0 = 0.02$ lies within the numerically stable region, close to the boundary of gradient stability.
- The growth rate $f(z)$ in the E_G estimator currently uses the GR approximation $f \approx \Omega_m(z)^{0.55}$, introducing a $\sim 0.3\%$ systematic (subdominant).

References

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